

Pairwise Sequence Alignment Method Using Flower Pollination Algorithm

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Abstract— Biomolecular sequence alignment is an important research area in bioinformatics. Based on Flower Pollination Algorithm (FPA), we have proposed a new sequence alignment algorithm, named Pairwise Sequence Alignment with Flower Pollination Algorithm (PSAFPA). PSAFPA helps in finding the optimal or near-optimal solution to sequence alignment problem. PSAFPA first finds the current best solution with the help of FPA and then divides this solution into various segments of non-matching regions to find the better results. These segments are rearranged using FPA and finally the best identity value of each segment is added into the current best solution to get the final alignment identity value. Such segmentation helped to get better results. We have also done comparison of PSAFPA with the well-known methods using various DNA sequence pairs, which resulted in to better performance of PSAFPA for a majority of sequence pairs.

Keywords— *Natural Computing algorithm; Sequence alignment; FPA; Lévy flight; DNA*

I. INTRODUCTION

Bioinformatics is the combination of biology, medicine, computer science and statistics. In bioinformatics, sequence alignment is an active research field. Sequence alignment involves the comparison of two or more sequences to get the maximum number of identical matches of residues (i.e. nucleotide bases for RNA and DNA, amino acids for protein). The unaligned and aligned sequences are shown in Fig. 1. Sequence alignment is an helps in solving various problems, such as finding homology, determining the origin of a sequence, finding specific sequences, protein structure predicting, constructing evolutionary (phylogenetic) trees.

Pairwise Sequence Alignment involves the comparison of two sequences, whereas more than two sequences are aligned at a time in multiple sequence Alignment. During sequence alignment, the aligned residues may result into a gap, match or mismatch. To handle insertions and deletions events we use Gaps (symbol '-') and in the scoring process they are assigned with penalties [1]. The alignment score for DNA can be simply found by assigning positive values to the similar residues in both the sequences and adding the values given to each

Unaligned Sequences

```
Seq.1:  CCCCACCTATTTTGT (16)
Seq.2:  TTGAGGTCTA (10)
```

Aligned Sequences

```
Seq.1:  CCC-CACCTAT-TTTTGT
Seq.2:  T--TGAGGTCTA-----
```

column. But in case of protein sequences, substitution matrices are used for scoring process, such as PAM [2] and BLOSUM [3].

Fig.1: Alignment of unaligned sequences.

Sequence alignment is of two types: global and local alignment [4], as shown in Fig. 2. Global sequence alignment tries to align the maximum residues from beginning to end for both the sequences. Local regions of highest similarity are identified within the two sequences regardless of their lengths in local sequence alignments.

```
(a) FTFTALILL-AVAV
    --FTAL-LLAAV--

(b) FTFTALILLAVAV
    F--TAL-LLA-AV
```

Fig. 2: (a) Local alignment (b) Global alignment.

Needleman-Wunsch algorithm which is Dynamic Programming (DP) based, is used to obtain optimal global alignment [5], whereas for finding optimal local alignment, Smith-Waterman algorithm is used [6]. These DP based algorithms produce optimal sequence alignment, but its

complexity increases when it comes to multiple sequence alignment. Various heuristic approaches were developed to decrease this complexity. For pairwise sequence alignment popularly used heuristic approaches are BLAST [7] and FASTA [8]. ClustalW, MUSCLE, T-Coffee package are often utilized heuristic approaches for multiple sequence alignment. Heuristics approaches are much faster but they do not provide optimal results. This is where the recently introduced Natural Computing Algorithms gain importance. They have the ability to provide optimal or near-optimal solutions. Some of these algorithms are: Genetic Algorithm(GA), Particle Swarm Optimization(PSO), Ant Colony Optimization(ACO), Artificial Bee Colony etc. In this paper we have developed a new algorithm using Flower Pollination Algorithm which is inspired from the pollination of flowering plant species.

S. R. Jangam, proposed a hybrid approach for pairwise alignment of DNA sequences using (ACO) and the simple GA [9]. The sequences pairs aligned by ACO are further processed by GA. It is capable of providing better results as compared to GA alone. It gave accurate results for small length sequences, but not so good for larger ones.

Gondro, C. used a GA-based approach for multiple sequence alignment, where the initial pairwise alignments was computed using the N-W algorithm [10]. This method performed better for sequences which are of medium or short lengths.

Lee, Z.J. proposed a technique for multiple sequence alignment by combining GA along with the ACO [11]. ACO is used to move out of local optima. GA is for providing the diversity of alignments. This approach was proposed to increase performance of GA. This approach shown better results when compared with relevant methods.

Garai, G. in their paper proposed a Genetic Algorithm based approach named CPAGA for finding the optimal alignment score for biomolecular sequences [12]. They reduced the search space by dividing the sequence pairs into various segments, to find the optimal solutions even for the larger sequences. Rubio-Largo introduced a Hybrid approach inspired by the foraging behaviour of honey bees for multiple sequence alignment [13]. It has also incorporated the Kalign method. This approach was compared with well-known methods and concluded to be a promising approach.

II. FLOWER POLLINATION ALGORITHM

Flower pollination deals with the transfer of pollen for pollination of plant species. Pollination is of two types biotic (global pollination) and abiotic (local pollination). 90% of pollination is biotic in which pollens are transferred by pollinators for example insects. As insects can fly large distances, thus to represent this we can use Levy flight. Only 10% of pollination is abiotic which is carried out with the help of wind or water for pollen transfer. Switch probability $P \in [0,1]$, is used to switch between Local pollination and global pollination, $P = 0.8$ works best in most cases [14].

Algorithmic representation of FPA:

1. Define the population of N flowers with random solutions
2. Define random variable $R \in [0,1]$ and switch probability $P \in [0,1]$.
3. While we reach maximum generations, for each flower in the population:
 - Draw a random variable R from $[0,1]$
 - If $R < P$, do global pollination
 - Else do local pollination
 - Evaluate the new solutions
 - Update new solutions in population, if they are better.
4. At the end of defined number of iterations find the global best solution.

III. PAIRWISE SEQUENCE ALIGNMENT WITH FLOWER POLLINATION ALGORITHM (PSAFPA)

In this Flower Pollination inspired approach, the main motive is to find optimal or near-optimal alignment with the different arrangements of gaps and residues in the sequences. Variable length DNA sequences are considered for the experiments. PSAFPA is represented algorithmically in Fig 6. The detailed approach is portrayed in the following steps:

1. Input two unaligned sequences. Sequences can be retrieved from public databases, each sequence have their own unique accession number.
2. Initialise the maximum number of generations G_{max} , the size of population N , random variable $R \in [0,1]$ and switch probability $P \in [0,1]$. Average number of generations are 2,000 for sequences of length 100–200, 10,000 generations for sequences of length 200–500.
3. Calculate length of both sequences. If sequences are of variable sizes, insert random gaps in the shorter sequence with respect to the longer sequence, such that length of both the sequences should be same after gaps insertion. Randomly generate the initial population of N alignments from the two input sequences.
4. Best solution is found from the initial population on the basis of objective function i.e. alignment-identity value. Objective function works on counting the identical residues in both the sequences as shown in Fig.3.

```

Initialise count to 0
for n in range (0, SeqLength):
  while Seq1[n]==Seq2[n] and Seq1[n]!="-":
    increment count by 1
  break
Identities= count

```

Fig. 3 Function for calculating the objective function.

5. Take the best solution from initial population and generate a new population by rearranging the residues by inserting gaps(5% of the length of bigger sequence) in both the sequences at the Indexes generated by FPA as shown in Fig.4. After this record identity scores for new solutions according to objective function and find the current best solution b^* .
6. Take the current best solution b^* and decompose it into segments of non-matching regions as shown in Fig.5. For each sequence pair of the segments, find their best alignment value s^* with the help of FPA. Summation of the best solution s^* of each segment to the current best solution b^* will give the final best solution.

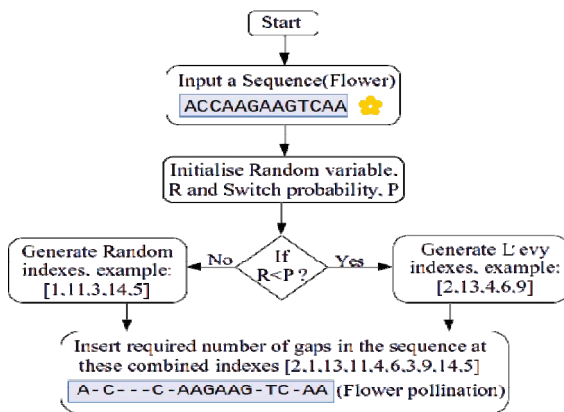


Fig. 4: FPA for generating Indexes to insert gaps.

Input Sequences :

Seq1: TACTTTTGTGCTCTTGGGGTCTCGTCTTCTACCCCTGGGGGATAATCAAGCCCCA
CGATAAACTCATCTTTGGAAAAGGAACCCGTGTGACTGTGGAACCAA

Seq2: TACTTTTGTGCTCTTGGGGAACTTTACGGGCCTCTCCGCCGGGGTGGGGGGCCTACAC
CGATAAACTCATCTTTGGAAAAGGAACCCGTGTGACTGTGGAACCAA

Current best solution b^*

Seq1: TACTTTTGTGCTCT-TGGGGTCT-CGTCTTCTTACCCCTGGGGATAATCAAGCCC-CA
Seq2: TACTTTTGTGCTCTTGGGAACTTTACGGGCCTCTCCGCCGGGGTGGGGGGGCCTACAC

Seq1: CGATAAACTCATCTTTGGAAAAGGAACCCGTGTGACTGTGGAACCAA
Seq1: CGATAAACTCATCTTTGGAAAAGGAACCCGTGTGACTGTGGAACCAA

Segmentation of non-matching regions

-T TG → T-TG GT AA → G-T-AA CGTCTT TACGGG → -CG-TCTT TACGGG- AC CT → AC-CT
 CTG GCC → -CTG-GC--C ATAATCAAG TGGGGGGGC → AT--AATC-A-AG-- -T-G--GG-GG-GGC CC-CA TACAC → --CC-CA T-A-CAC-

IV. RESULTS AND DISCUSSION

Performance of proposed PSAFPA has been evaluated using DNA sequence. We have retrieved 12 pairs of DNA sequences for the experiment from the database of NCBI. Variable length DNA sequences are taken for experiment.

PSAFPA performance is compared with the relevant sequence alignment methods. Alignment identities produced by PSAFPA have been compared with DP(Dynamic Programming) and BLAST. The identity scores are shown in Table 1. In Table 1, the values within round brackets shows the identity percentage. Calculation of identity percentage is given below:

$$I\% = \left[\frac{(2 \times S)}{(L_1 + L_2)} \right] \times 100 \quad (2)$$

where S is the number of identities found in 2 Sequences, L_1 and L_2 are the lengths of input sequences.

From Table I, it can be seen that PSAFPA produced optimal or near-optimal results as compared to other approaches. Highest identity values are shown with the bold values.

It is notified that division of the best solution produced with the help of FPA into various segments (sequence pairs) of non-matching regions and again applying FPA on these segments gave better results. From Fig.7, we observed that the better results were achieved with the help of segmentation.

Fig. 5: Decomposition of current best solution b^* into segments of non-matching regions and rearrangement of segments using FPA.

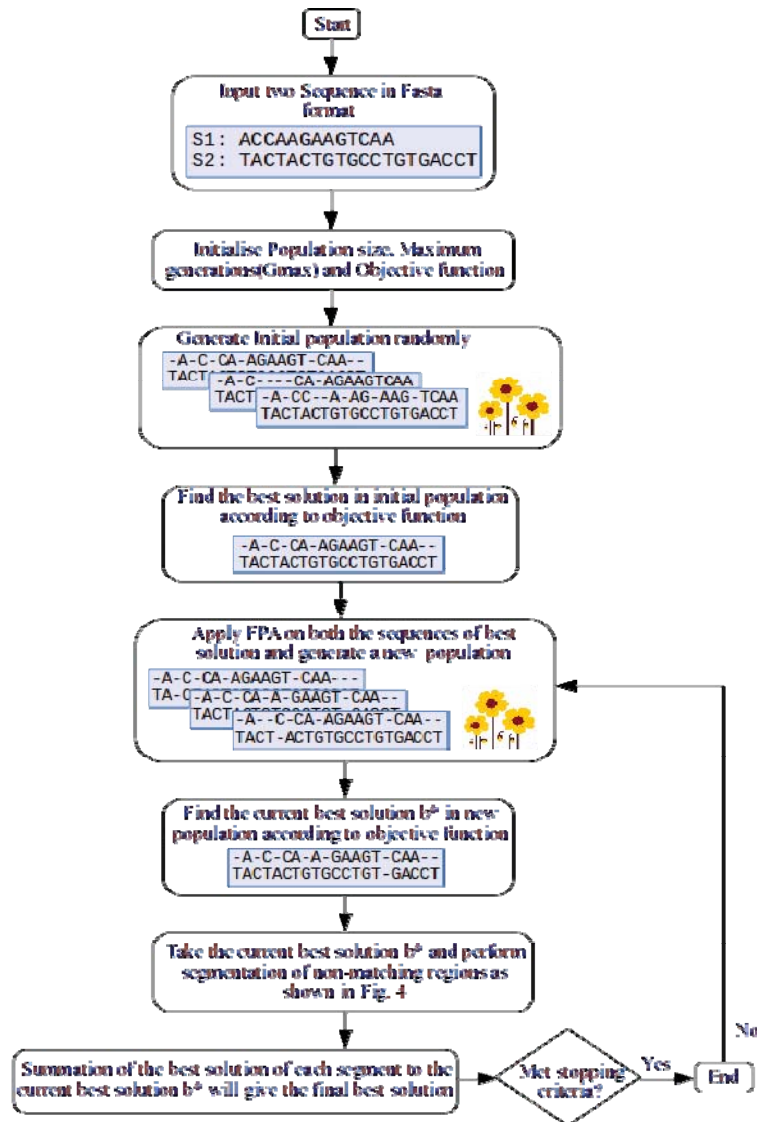


Fig. 6: Flowchart of the proposed PSAFPA algorithm.

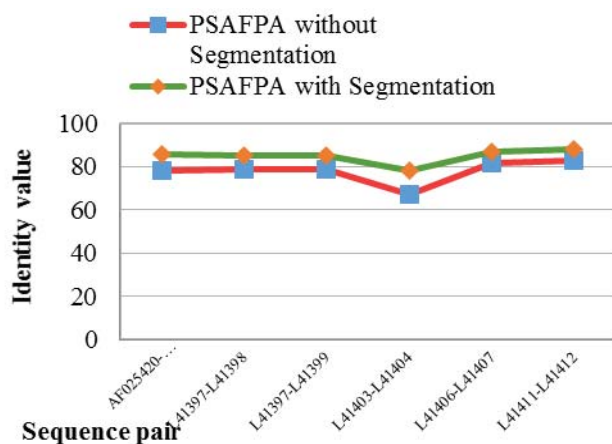


Fig. 7 : Comparison of the PSAFPA results for DNA sequence pairs, in terms of identity values with and without segmentation.

Figure 8 represents the comparison of 3 approaches in percentage of identity scores. The sequence pairs for which percentage of identity produced by 3 approaches is same, were not taken for comparison. From Fig.8, it can be seen that the percentage identity scores of PSAFPA are better than DP and BLAST for a majority of sequence pairs.

Table I: Comparison of results produced by PSAFPA with well known approaches.

Accession Number	Sequence length	BLAST	DP	PSAFPA
AF025420	103	47	83	86
AF025417	106	(44.98%)	(79.43%)	(82.30%)
L41396	102	91	91	91
L41398	102	(89.22%)	(89.22%)	(89.22%)
L41397	102	59	83	85
L41398	102	(57.84%)	(81.37%)	(83.33%)
L41397	102	56	83	85
L41399	108	(53.33%)	(79.05%)	(89.95%)
L41400	96	85	85	85
L41401	93	(89.95%)	(89.95%)	(89.95%)
L41403	93	56	81	78
L41404	105	(56.57%)	(81.82%)	(78.79%)
L41406	105	68	89	87
L41407	99	(66.67%)	(87.25%)	(85.29%)
L41407	99	99	99	99
L41408	99	(100%)	(100%)	(100%)
L41411	96	62	87	88
L41412	99	(63.59%)	(89.23%)	(90.26%)
AV239402	657	No-significant similarity	314	315
AV239646	661		(47.65%)	(47.80%)
AB023276	457	453	453	453
AB023278	457	(99.12%)	(99.12%)	(99.12%)
AB023276	457	456	456	456
AB023279	457	(99.78%)	(99.78%)	(99.78%)

V. CONCLUSIONS

The PSAFPA method is inspired from the pollination of flowering species, to solve pairwise sequence alignment problem. Traditional approaches already exist for solving such problems, which also provide optimal solutions, but they become complex and demands more computer resources when it comes to multiple sequence alignment. A nature inspired computing algorithm may bring about less complex solutions and produce optimal or near-optimal results. The proposed PSAFPA algorithm does not use any complex matrix for sequence alignment as required by DP. The proposed algorithm is much flexible to introduce new gaps and rearrange the positions of residues and gap in an alignment, which helps to find a better alignment. The main purpose is to find the solution for sequence alignment problem and to compare its results with well-known approaches. Simple alignment identity value is used as the objective function. It is observed from the results that the proposed algorithm produced better alignment results for a majority of sequence pairs when compared with DP and BLAST. This research work can be further implemented for aligning multiple sequences.

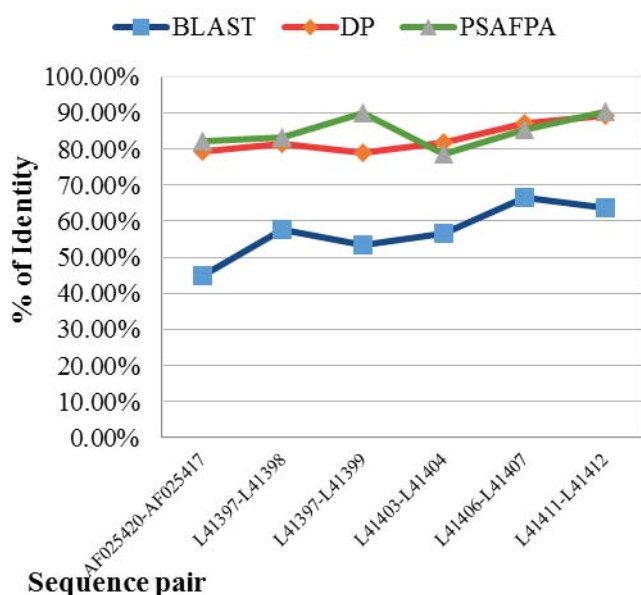


Fig. 8: Comparison of PSAFPA results for DNA sequence pairs, in terms of identity percentage with BLAST and DP.

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